

WTW 158 Worksheet 10 on Integration

A. TEXTBOOK PROBLEMS

Integral and Area problems:

p 375 no 4

p 388 no 33, 37, 39, 51

p 399 no 45, 53, 55

Integration problems:

p 399 no 19, 25, 29, 39, 43,

p 408 no 9 - 17 (odd)

p 418 no 29 - 45 (odd), 63, 65, 69)

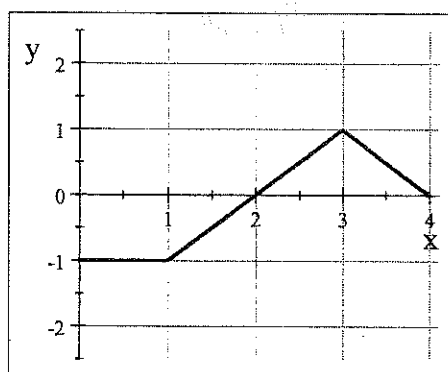
B. MORE PROBLEMS

1. 55 Integration Challenge, available on ClickUP under Worksheets

2. Technique Mastering Problems p 49 N - Integration by inspection

C. TEST QUESTIONS

1.



For the function f on the interval $[0, 4]$, of which the graph is shown:

1.1 $\int_0^4 f(x) dx =$

(a)	-0.5	(b)	-1.5	(c)	2.5	(d)	0.5
(e)	3	(f)	None of these				

1.2. The area enclosed between the curve and the x-axis on $[0, 4]$ is given by

(a)	-0.5	(b)	-1.5	(c)	2.5	(d)	0.5
(e)	3	(f)	None of these				

2. Find the area bounded by the curve $y = 2x - x^2$, the x-axis and the lines $x = -1$ and $x = 2$.

D. SELECTED ANSWERS

A. Integral and area problems p 375 no 4 $RHS = \frac{\pi}{8} (\sin \frac{\pi}{8} + \frac{1}{\sqrt{2}} + \sin \frac{3\pi}{8} + 1)$,

$$LHS = \frac{\pi}{8} (0 + \sin \frac{\pi}{8} + \frac{1}{\sqrt{2}} + \sin \frac{3\pi}{8})$$

p 388 See p A93; p 399 See p A93

Integration problems p399 See p A93; p 408 See p A94; p 418 See p.A94

B. 1. See ClickUP 2. See ClickUP

C. 1.1 [a] 1.2 2.5 2. $\frac{8}{3}$

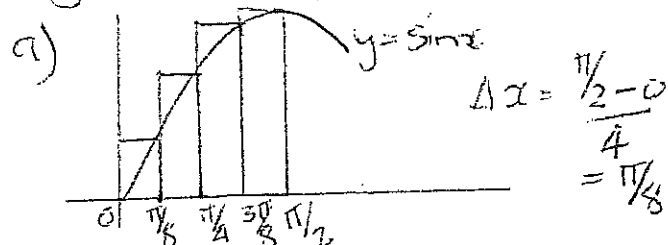
Worksheet 10 (Integration)

A Textbook Problems

Integral and area Problems

p 375 no 4

$$y = \sin x, x \in [0, \frac{\pi}{2}]$$



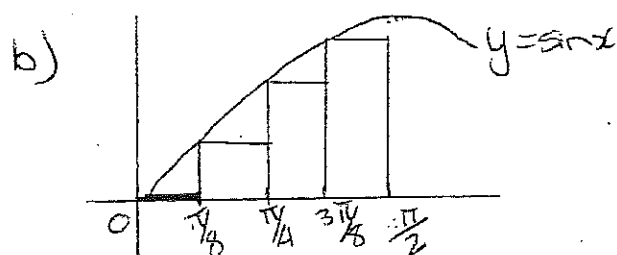
$$RHS = \frac{\pi}{8} [f(\frac{\pi}{8}) + f(\frac{\pi}{4}) + f(\frac{3\pi}{8}) + f(\frac{\pi}{2})]$$

$$= \frac{\pi}{8} [\sin \frac{\pi}{8} + \sin \frac{\pi}{4} + \sin \frac{3\pi}{8} + \sin \frac{\pi}{2}]$$

$$= \frac{\pi}{8} [\sin \frac{\pi}{8} + \frac{1}{\sqrt{2}} + \sin \frac{3\pi}{8} + 1]$$

f is increasing on

RHS is an overestimate



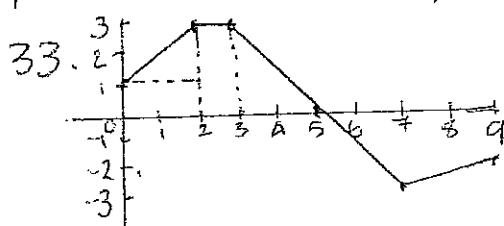
$$LHS = \frac{\pi}{8} [\sin 0 + \sin \frac{\pi}{8} + \sin \frac{\pi}{4} + \sin \frac{3\pi}{8}]$$

$$= \frac{\pi}{8} [0 + \sin \frac{\pi}{8} + \frac{1}{\sqrt{2}} + \sin \frac{3\pi}{8}]$$

f is increasing on $[0, \frac{\pi}{2}]$

LHS is an underestimate

p 388 no 33, 37, 39, 51



$$a) \int_0^2 f(x) dx = (\frac{1}{2} \times 2 \times 2) + (2 \times 1) = 4$$

①

$$b) \int_0^5 f(x) dx = \int_0^2 f(x) dx + \int_2^3 f(x) dx + \int_3^5 f(x) dx = 4 + (1 \times 3) + (\frac{1}{2} \times 2 \times 3) = 10$$

37. $\int_{-3}^0 (1 + \sqrt{9-x^2}) dx$

$$= \frac{1}{4} \text{ area of } \odot + \text{area } \square$$

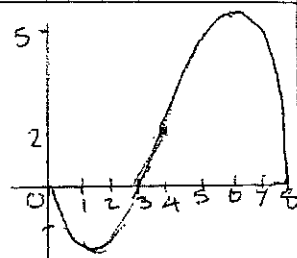
$$= \frac{1}{4} (\pi \times 3^2) + (3 \times 1) = 3 + \frac{9\pi}{4}$$

39. $\int_{-4}^3 |2x| dx$

$$= (\frac{1}{2} \times 4 \times 2) + (\frac{1}{2} \times 3 \times \frac{3}{2}) = 4 + \frac{9}{4} = \frac{25}{4}$$

51.

Smallest to largest



$$\int_0^3 f(x) dx < f'(1) < \int_0^3 f(x) dx < \int_4^8 f(x) dx < \int_3^8 f(x) dx$$

p 399 no 45, 53, 55

45. $\int_0^4 \sqrt{x} dx$ (= area)

$$= \frac{2x^{3/2}}{3} \Big|_0^4 = \frac{2}{3} (8) - 0 = \frac{16}{3}$$

53. $\int_{-1}^2 x^3 dx = \frac{x^4}{4} \Big|_{-1}^2$

$$= \frac{1}{4} (16 - 1) = \frac{15}{4}$$

55. $\int_{-2}^1 x^{-4} dx = \frac{x^{-3}}{-3} \Big|_{-2}^1 = -\frac{3}{8}$

$f(x) = x^{-4} = \frac{1}{x^4}$ is discontinuous on $[-2, 1]$. FTC 2 cannot be applied

Integration Problems

P 399 no 19, 25, 29, 39, 43

P 408 no 9-17 odd

P 418 no 29-4 odd
no 63-69 odd

P 399

$$19. \int_1^3 (x^2 + 2x - 4) dx$$

$$= \left(\frac{x^3}{3} + x^2 - 4x \right) \Big|_1^3 = \frac{26}{3}$$

$$25. \int_{\pi/6}^{\pi} \sin e \, de = [\cos e] \Big|_{\pi/6}^{\pi}$$

$$= -\cos \pi + \cos \pi/6$$

$$= -(-1) + \frac{\sqrt{3}}{2}$$

$$29. \int_1^4 \frac{2+x^2}{\sqrt{x}} dx = \int_1^4 \left(\frac{2}{\sqrt{x}} + \frac{x^2}{\sqrt{x}} \right) dx$$

$$= \left(4x^{1/2} + \frac{2}{5}x^{5/2} \right) \Big|_1^4$$

$$= [4(2) + \frac{2}{5}(32)] - [4 + \frac{2}{5}]$$

$$= \frac{82}{5}$$

$$39. \int_{-1}^1 e^{u+1} du = e^{u+1} \Big|_{-1}^1$$

$$= e^2 - e^0 = e^2 - 1$$

$$43. f(x) = \begin{cases} \sin x & \text{if } 0 \leq x \leq \pi/2 \\ \cos x & \text{if } \pi/2 \leq x \leq \pi \end{cases}$$

$$\int_0^{\pi} f(x) dx = \int_0^{\pi/2} \sin x dx + \int_{\pi/2}^{\pi} \cos x dx$$

$$= (-\cos x) \Big|_0^{\pi/2} + \sin x \Big|_{\pi/2}^{\pi}$$

$$= (0 + 1) + (0 - 1) = 0$$

P 408

$$9. \int v(v^2 + 2)^2 dv$$

$$= \int (v^5 + 4v^3 + 4v) dv$$

$$= \frac{v^6}{6} + v^4 + 2v^2 + C$$

$$11. \int \frac{1 + \sqrt{x} + x}{x} dx$$

$$= \int \left(\frac{1}{x} + \frac{1}{\sqrt{x}} + 1 \right) dx$$

$$= \ln|x| + 2\sqrt{x} + x + C$$

$$13. \int (\operatorname{cosec}^2 t - 2e^t) dt$$

$$= -\cot t - 2e^t + C$$

$$15. \int (2 + \tan^2 \theta) d\theta$$

$$= \int 2 + (\sec^2 \theta - 1) d\theta$$

$$= \int (1 + \sec^2 \theta) d\theta = \theta + \tan \theta + C$$

$$17. \int 2^t (1 + 5^t) dt$$

$$= \int (2^t + 2^t 5^t) dt$$

$$= \int 2^t + 10^t dt$$

$$= \frac{2^t}{\ln 2} + \frac{10^t}{\ln 10} + C$$

P 418

$$29. \int 5^t \sin(5^t) dt$$

$$= \frac{-\cos(5^t)}{\ln 5} + C$$

$$31. \int \frac{(\arctan x)^2}{x^2 + 1} dx$$

$$= \frac{1}{3} (\arctan x)^3 + C$$

$$33. \int \cos(1 + 5t) dt$$

$$= \frac{1}{5} \sin(1 + 5t) + C$$

$$35. \int (\sqrt{\cot x}) (\operatorname{cosec}^2 x) dx$$

$$= \int (\cot x)^{1/2} x \operatorname{cosec}^2 x dx$$

$$= -\frac{2}{3} (\cot x)^{3/2} + C$$

$$37. \int \sinh^2 x \cosh x dx$$

$$= \frac{1}{3} \sinh^3 x + C$$

$$39. \int \frac{\sin 2x}{1 + \cos^2 x} dx$$

$$= 2 \int \frac{\sin x \cos x}{1 + \cos^2 x} dx$$

$$= -\ln|1 + \cos^2 x| + C$$

$$41. \int \cot x \, dx = \int \frac{\cos x}{\sin x} \, dx = \ln |\sin x| + C$$

$$43. \int \frac{dx}{(\sqrt{1-x^2})(-\sin^{-1}x)} = \ln |\sin^{-1}x| + C$$

$$45. \int \frac{1+x}{1+x^2} \, dx = \int \frac{1}{1+x^2} \, dx + \int \frac{x}{1+x^2} \, dx = \tan^{-1}x + \frac{1}{2} \ln |1+x^2| + C$$

$$63. \int_0^{13} \frac{dx}{\sqrt[3]{(1+2x)^2}} = \int_0^{13} (1+2x)^{-2/3} \, dx = \frac{3x}{1/2} (1+2x)^{1/3} \Big|_0^{13} = \frac{3}{2} \left[(1+26)^{1/3} - 1^{1/3} \right] = \frac{3}{2} [3 - 1] = 3$$

$$65. \int_0^a x \sqrt{x^2+a^2} \, dx = \frac{1}{2} \times \frac{2}{3} (x^2+a^2)^{3/2} \Big|_0^a = \frac{1}{3} \left[(2a^2)^{3/2} - (a^2)^{3/2} \right] = \frac{1}{3} \left((2)^{3/2} + 1 \right) (a^3)$$

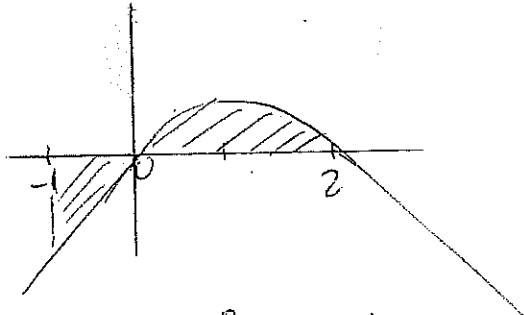
$$67. \int_e^{e^4} \frac{dx}{x \sqrt{\ln x}} = \frac{2}{1} (\ln x)^{1/2} \Big|_e^{e^4} = 2 (\sqrt{\ln e^4} - \sqrt{\ln e}) = 2 (\sqrt{4 \ln e} - 1) = 2 (2 - 1) = 2$$

C. Test Questions

$$1.1 \int_0^4 f(x) \, dx = \left(1 + \frac{1}{2}\right) + 1 = \frac{3}{2} \quad [a]$$

$$1.2. \text{Area} = \int_0^4 |f(x)| \, dx = \frac{3}{2} + 1 = \frac{5}{2} = 2.5$$

$$2. \quad y = 2x - x^2, \quad x \in [-1, 2]$$



$$\begin{aligned} \text{Area} &= -\int_{-1}^0 (2x - x^2) \, dx + \int_0^2 (2x - x^2) \, dx \\ &= -\left(x^2 - \frac{x^3}{3}\right) \Big|_{-1}^0 + \left(x^2 - \frac{x^3}{3}\right) \Big|_0^2 \\ &= -\left[0 - \left(1 + \frac{1}{3}\right)\right] + \left[\left(4 - \frac{8}{3}\right) - 0\right] \\ &= \frac{4}{3} - \frac{8}{3} + 4 = \frac{-4}{3} + 4 \\ &= \frac{-4}{3} + \frac{12}{3} = \frac{8}{3} \end{aligned}$$